FRONTEND DOCUMENTATION

1. Overview

The frontend is a web-based user interface that allows users to:

- Enter exoplanet parameters
- Send data to backend API
- View habitability predictions

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2. Technologies Used

- HTML
- CSS
- JavaScript
- Bootstrap (for styling)

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3. Folder Structure

```
frontend/
├── index.html
├── style.css
└── script.js
```

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4. UI Design

The UI consists of:

- Input form for planetary parameters
- Submit button
- Result display section

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5. Input Form

The form collects:

- Planet Radius
- Planet Mass
- Orbital Period
- Stellar Temperature

Each field:

- Uses proper labels
- Has placeholder values
- Accepts numeric input only

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6. Frontend–Backend Integration

- JavaScript uses **Fetch API**
- Sends POST request to `/predict`
- Data is sent in JSON format

Example Request:

```
fetch("http://127.0.0.1:5000/predict", {
    method: "POST",
    headers: {
        "Content-Type": "application/json"
    },
    body: JSON.stringify(data)
});
```

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7. Displaying Results

Results are shown using:

- Bootstrap cards for success
- Alerts for errors

Displayed information:

- Habitability status
- Score
- Confidence

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8. Error Handling

Frontend handles:

- Empty inputs
- Invalid numbers
- Backend errors
- Connection failures

Example:

- If backend is down → "Cannot connect to backend"
- If invalid input → "All fields must be valid numbers"

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9. Responsiveness

- Built using Bootstrap grid system
- Works on:
  - Desktop
  - Tablet
  - Mobile

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10. Integration Testing

Flow tested:

```
User Input → Frontend → Backend API → ML Model → Response → UI Display
```

All scenarios verified:

- Valid input
- Invalid input
- Server errors

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11. Conclusion

The frontend provides a simple and interactive interface that successfully integrates with the backend API and displays predictions effectively.